GRAPH THEORY:
HOMEWORK 9
BOOK SECTION 4.2 PART II AND 5.1.

ROB TOMPKINS

1. Prove the part of the hypercube Q_k is k-connected by constructing k pairwise internally disjoint x, y-paths for each vertex pair $x, y \in V(Q_k)$. (Hint: Use induction to show that for all k, for all $x, y \in V(Q_k)$ there are k internally disjoint x, y-paths. Why does this suffice?)

Proof. We prove by induction on $k \geq 1$ that every two distinct vertices of Q_k are joined by k pairwise internally disjoint paths. The case $k = 1$ holds because $Q_1 = K_2$.

Suppose $k \geq 2$ and the assertion holds for Q_{k-1}. View the vertices as binary k-tuples, with adjacency given by changing one coordinate. Let $x \neq y$.

Case 1: x and y agree in some coordinate j. The vertices having their common value in coordinate j induce a copy H of Q_{k-1}. By induction, H contains $k-1$ pairwise internally disjoint x, y-paths. The vertices having the other value in coordinate j induce another copy H' of Q_{k-1}. Let x' and y' be obtained from x and y by changing coordinate j. Since H' is connected, choose an x', y'-path R in H'. The path formed by xx', R, and $y'y$ has all its internal vertices in H'. It is therefore internally disjoint from the $k-1$ paths in H, giving the required k paths.

Case 2: x and y differ in every coordinate. For $j \in \{1, \ldots, k\}$, form a path P_j by starting at x and changing coordinates in the cyclic order

$$j, j+1, \ldots, k, 1, \ldots, j-1.$$

Each coordinate is changed exactly once, so P_j is a path ending at y. At an internal vertex after t steps, where $1 \leq t \leq k-1$, the coordinates that differ from x are precisely the cyclic interval of length t beginning at j. This is a nonempty proper cyclic interval, and its starting coordinate is uniquely determined: it is the coordinate in the interval whose cyclic predecessor is outside the interval. Thus an internal vertex determines both t and j, and cannot lie on two different paths P_j. These k paths are pairwise internally disjoint.

This completes the induction. Theorem 4.2.21 (the global form of Menger's Theorem) now implies that Q_k is k-connected. Explicitly, if fewer than k vertices are deleted and x, y survive, each deleted vertex belongs internally to at most one of our k paths, so at least one x, y-path survives. Also $|V(Q_k)| = 2^k \geq k+1$, as required for k-connectivity. □

2. Let G be a k-connected graph, and let S, T be disjoint subsets of $V(G)$ with $|S| \geq k$ and $|T| \geq k$. Prove that G has a k pairwise disjoint S, T-paths. (Hint: Consider adding new vertices and try to use Menger's Theorem.)

Proof. Assume $k \geq 1$; for $k = 0$ the assertion is vacuous. Construct H from G by adding distinct new vertices a and b, joining a to every vertex of S and b to every vertex of T. Do not add an edge ab.

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We claim that no set of fewer than k vertices separates a from b in H. Indeed, let $U \subseteq V(H) \setminus \{a, b\}$ with $|U| < k$. Since $|S|, |T| \geq k$, choose $s \in S \setminus U$ and $t \in T \setminus U$. Since G is k-connected, $G - U$ is connected, so it contains an s, t-path. Together with as and tb, this gives an a, b-path in $H - U$, proving the claim.

By Menger's Theorem 4.2.17, since a, b are nonadjacent, H has k pairwise internally disjoint a, b-paths. Delete a and b from each path. The resulting paths lie in G, start in S, and end in T. They are pairwise vertex-disjoint, including their endpoints, because all of their vertices were internal vertices of the original paths.

If an S, T-path is required to have no internal vertices in $S \cup T$, shorten each resulting path as follows. Orient it from its S-endpoint to its T-endpoint; take the first vertex of T on it and the last vertex of S preceding that vertex. The subpath between these vertices has no internal vertex in $S \cup T$. Taking subpaths preserves pairwise vertex-disjointness, so these are the required k disjoint S, T-paths. □

3. Let G be a graph whose odd cycles are pairwise intersecting, meaning that every two odd cycles in G have a common vertex. Prove $\chi(G) \leq 5$. (Hint: Try the contrapositive. If $\chi(G) \geq 6$, consider an optimal coloring and the subgraphs induced by groups of its groups of color classes. Recall that a graph with no odd cycle is bipartite.)

Proof. We prove the contrapositive. Suppose $r = \chi(G) \geq 6$ and choose an optimal proper coloring with color classes $C_1, \ldots, C_r$ (Definition 5.1.4). Let

$$A = C_1 \cup C_2 \cup C_3, \qquad B = C_4 \cup C_5 \cup C_6.$$

The induced graph $G[A]$ is not bipartite: if it were, we could replace the three colors used on A by two colors, keeping colors $4, \ldots, r$ on the other vertices. This would be a proper $(r-1)$-coloring of G, contrary to the definition of r. The same argument shows that $G[B]$ is not bipartite: replace its three colors by two while retaining all other colors.

A graph is bipartite if and only if it contains no odd cycle, the characterization used in Examples 5.1.3 and 5.1.5. Hence $G[A]$ and $G[B]$ each contain an odd cycle. Since $A \cap B = \varnothing$, these two odd cycles have no common vertex. Thus $\chi(G) \geq 6$ implies the existence of two vertex-disjoint odd cycles. Taking the contrapositive proves $\chi(G) \leq 5$ under the stated hypothesis. □

4. Prove that a graph G is m-colorable if and only if $\alpha(G \square K_m) \geq n(G)$.

Proof. Let $m \geq 1$ and identify $V(K_m)$ with $[m] = \{1, \ldots, m\}$. By Definition 5.1.9, the vertices of $G \square K_m$ are pairs (v, i), and two distinct pairs $(u, i), (v, j)$ are adjacent exactly when

$$u = v \text{ and } i \neq j, \qquad \text{or} \qquad uv \in E(G) \text{ and } i = j.$$

Suppose first that $c : V(G) \to [m]$ is a proper coloring. Set

$$I = \{(v, c(v)) : v \in V(G)\}.$$

This set contains exactly one pair for each v, so $|I| = n(G)$. Two distinct elements of I have different first coordinates. If their first coordinates are adjacent in G, their second coordinates differ because c is proper. Thus neither adjacency condition in the product holds, and I is independent. Consequently $\alpha(G \square K_m) \geq n(G)$.

Conversely, suppose $G \square K_m$ has an independent set I of size at least $n(G)$. For each $v \in V(G)$, the fiber

$$F_v = \{(v, i) : i \in [m]\}$$

induces a clique K_m. Therefore I contains at most one vertex from each F_v. There are $n(G)$ such fibers, so $|I| = n(G)$ and I contains exactly one vertex $(v, c(v))$ from every fiber. This defines a map $c : V(G) \to [m]$. If $uv \in E(G)$ and $c(u) = c(v) = i$, then (u, i) and (v, i) would be adjacent vertices of I, a contradiction. Thus c is a proper m-coloring of G. This is the correspondence between color classes and independent sets described in Remark 5.1.2. □

5. Prove that every graph G has a vertex ordering relative to which greedy coloring uses $\chi(G)$ colors.

Proof. Let $r = \chi(G)$ and choose an optimal proper coloring $f : V(G) \to \{1, \ldots, r\}$. Order all vertices of color 1 first, then those of color 2, and so on through color r, with any order within each color class. Apply the greedy coloring of Algorithm 5.1.12 to this ordering, and call the resulting coloring g.

We prove by induction along the ordering that $g(v) \leq f(v)$ for every vertex v. Suppose this has been established for all vertices preceding v, and put $i = f(v)$. Any earlier neighbor u of v satisfies $f(u) \leq i$ by the ordering. In fact $f(u) < i$, since vertices of the same color class are nonadjacent (Remark 5.1.2). By induction,

$$g(u) \leq f(u) \leq i - 1.$$

Thus color i is absent from all earlier neighbors of v. Since greedy coloring chooses the least available positive integer, $g(v) \leq i = f(v)$. This argument also establishes the base case, where there are no earlier neighbors.

Therefore greedy coloring uses at most r colors. Its output is a proper coloring, so it cannot use fewer than $\chi(G) = r$ colors. It consequently uses exactly $\chi(G)$ colors. For the empty graph, the empty ordering has the same property, with zero colors. □